

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment & Information Technology

Department of Computer Science

COS326 - Database Systems

Semester Test Paper

Total Marks: 100

28 August 2025

Time/Duration: 12:30pm - 14:00pm (1 Hour 30 Minutes)

Examiner: Mr. S.M. Makura

External Moderator: Prof Sonia Berman (UCT)

Initials and Surname: _____

Student Number: _____

Degree: _____

Instructions

1. Fill in the semester test paper with your details above.
2. Read the question paper carefully and answer all the questions.
3. This paper comprises of 26 questions on 29 pages. It consists of three sections,

namely Section A (Multiple Choice & Fill in the blank questions) - 20 marks, Section B (Application and Theory Based Questions) - 40 Marks, Section C (Scenario Based questions) - 40 Marks.

4. Answer **all** the questions in the answer book provided. After you are done, submit **both** the semester test paper together with your answer book.
5. You have 90 minutes to complete this semester test. Spend approximately the following minutes per each section:
   - Section A - 15 minutes
   - Section B - 30 minutes
   - Section C - 45 minutes
6. This is a closed book examination: you may therefore not have any study material with you.
7. Please switch off your smartphone/tablet and keep it off for the duration of the semester test.
8. All examination regulations and 'code of conduct' of the University of Pretoria are applicable during this semester test.

Integrity Statement:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Section A: Multiple Choice and Fill in the Blanks Questions (20 Marks) (Spend approximately 15 minutes in this section)

Question 1

1. Which limitation of RDBMSs is most directly addressed by the use of Object Identifiers (OIDs) in OODBMSs?

A. Limited operations
B. Semantic overloading
C. Poor representation of real-world entities
D. Homogeneous data structure

Question 2

Nothando writes `SELECT c FROM Customer c JOIN c.orders o WHERE o.total > 500`. What does this query do?

A. Retrieves customers with more than 500 orders
B. Retrieves customers who placed orders over 500
C. Retrieves orders with totals over 500
D. Retrieves all customers regardless of order total

Question 3

You want to update a product's price in JPQL. Which query is correct?

A. `UPDATE Product SET price = 200 WHERE id = 1`
B. `MODIFY Product SET price = 200 WHERE id = 1`
C. `SET Product.price = 200 WHERE Product.id = 1`
D. `UPDATE Product p SET p.price = 200 WHERE p.id = 1`

Question 4

You want to store a full address as a single column in a table. Which SQL:2016 feature allows this?

A. Row type
B. ENUM type
C. Composite type
D. Column type

Question 5

A financial analytics firm stores client portfolios using a structured UDT called `PortfolioType`, which includes an array of asset values and a method `sumAssets()` that sums them. The firm wants to allow inheritance for specialized portfolios like `RetirementPortfolio` and `GrowthPortfolio`.

Which combination of clauses ensures both instantiation and subtype extension?

A. FINAL, INSTANTIALBLE

B. NOT FINAL, ABSTRACT

C. NOT FINAL, INSTANTIABLE

D. SUBTYPE, INSTANTIABLE

Question 6

Which of the following statements about triggers in PostgreSQL is FALSE?

A. A BEFORE trigger executes before the specified operation.

B. Triggers are automatically inherited by subtables.

C. Trigger functions can be reused across multiple triggers.

D. An AFTER trigger executes after the specified operation.

Question 7

A hospital database stores patient records. The system must automatically log any deletion of a patient record into an audit table.

Which PostgreSQL trigger timing should be used?

A. BEFORE DELETE
B. AFTER DELETE
C. INSTEAD OF DELETE
D. ON DELETE CASCADE

Question 8

Which of the following statements is true about the root element in an XML document?

A. There can be multiple root elements

B. Only elements with attributes can be root

C. An XML document must have exactly one root element

D. The root element must be named 'root'

Question 9

An airline stores flight data in XML. Each `<flight>` element contains `<destination>`, `<departureTime>`, and `<status>`. You want to list destinations of flights that depart after 18:00. Later, you will filter by status separately.

Which XPath expression correctly retrieves destinations of flights departing after 18:00?

A. `//flight[departureTime > "18:00"]/destination`
B. `//flight[destination > "18:00"]/departureTime`
C. `//flight[status = "on-time"]/departureTime > "18:00"`
D. `//flight[departureTime > "18:00"]/status = "on-time"`

Question 10

An organization stores employee data in XML. Each `<employee>` has `<name>`, `<role>`, and may contain nested `<employee>` elements for subordinates. You want to extract names of all subordinates under a manager named "Pholoso".

Which XPath expression is suitable?

A. `//employee[name="Pholoso"]/employee/name`

B. `//employee[name="Pholoso"]/name`

C. `//employee/employee[name="Pholoso"]/name`

D. `//employee[name="Pholoso"]/role="manager"`

Question 11

A JPA entity class is a _____ class, i.e. an ordinary Java class that is marked (annotated) as having the ability to represent objects in the database.

Question 12

Fill in the blanks to create a `TypedQuery` for retrieving all departments using JPQL:

```
TypedQuery _________ query = em.createQuery("SELECT d FROM
Department d", Department.class);
```

Question 13

Given an EntityManager, `em`, you begin a transaction in the following way (write the full line of code including the semicolon at the end).

Question 14

The _____ in SQL:2016 allows access to attributes of referenced objects directly in queries.

Question 15

The _____ loop in PL/pgSQL is used to iterate over elements in an array.

Question 16

The _____ clause in SQL:2016 creates a unique object identifier for each row in a typed table

Question 17

True/False: In PostgreSQL, table inheritance automatically enforces foreign key constraints across parent and child tables.

Question 18

An XML schema is commonly referred to as XSD, which stands for _____

Question 19

In XML Schema, the type `xs:_____` is used to define elements that contain textual data.

Question 20

True/False: XPath is a language used to transform XML documents.

Section B Application and Theory - Based Questions (40 Marks)

(Spend approximately 30 minutes in this section)

Question 21 (15 Marks)

Object DBMS

(a) Evaluate the pros and cons of embedding OODB constructs directly into host languages versus extending SQL with OO features. (4 marks)

(b) Forensic investigators log evidence over several days. You need to support long-lived transactions. The following code snippet begins and commits a transaction:

```
EntityTransaction tx = em.getTransaction();
tx.begin();
// persist multiple EvidenceItem objects
tx.commit();
```

Modify the above snippet to include rollback logic in case of failure. (6 marks)

(c) Briefly explain what a Semantic Information Manager is and how it differs from traditional relational or object databases. (5 marks)

Question 22 (15 Marks)

ORDBMS

Study the following SQL statements and answer the following questions:

```
CREATE TABLE transactions (
    account_id serial PRIMARY KEY,
    changes numeric[]
);

INSERT INTO transactions (changes) VALUES
    (ARRAY[100, -50, 200, -30]),
    (ARRAY[-10, 20, 30]),
    (ARRAY[50, 100, -25, -25]);
```

You have executed the above statements in PostgreSQL, and they ran successfully.

(a) Complete the function to sum only positive changes.

```
CREATE _________ sum_positive_changes(changes numeric[])
RETURNS NUMERIC AS $$

__________
    total numeric := 0;
    val numeric;
BEGIN
    _______ ____ IN ARRAY changes
    ________
        IF val > 0 THEN
            _________;
        END IF;
    END ________;

    RETURN ________;
END;
$$ _____________;
```

(10marks)

(b) Write the select statement to display `account_id` and `sum` of positive changes. (1 Marks)

(c) Compare the limitations of traditional RDBMSs with the extended capabilities of ORDBMSs like PostgreSQL. Focus on support for complex types, inheritance, and procedural logic. (5 Marks)

Question 23 (10 Marks)

Semi structured and XML

You are working with an XML dataset that contains information about customer orders. Each <order> element includes details such as the order ID, customer name, total amount, and shipping status. Below is a sample of the XML structure:

```
<orders>
    <order>
        <order_id>201</order_id>
        <customer>Emily Clark</customer>
        <total>250</total>
        <status>Shipped</status>
    </order>
    <order>
        <order_id>202</order_id>
```

```
        <customer>Michael Lee</customer>
        <total>150</total>
        <status>Pending</status>
    </order>
    <order>
        <order_id>203</order_id>
        <customer>Sarah Kim</customer>
        <total>400</total>
        <status>Shipped</status>
    </order>
</orders>
```

Using XPath queries, answer the following questions:

(a) Write an XPath query to retrieve all customers with shipped orders (3 Marks)

(b) Write an XPath query to retrieve the name of the customer with order id 203 (3 Marks).

(c) With the aid of XML code examples, differentiate between the simple and complex definition types. (4 Marks)

Section C: Scenario Based Questions (40 Marks)

(Spend approximately 45 minutes in this section)

Question 24 (15 Marks)

OODBMS

Study the following scenario and answer the following questions:

> Jacob is developing a ride-hailing app using ObjectDB. The ride-hailing app tracks rides and each ride has:
>
> - `driverName: String`.
> - `passengerName: String`.
> - `distance: Double` (km travelled).
> - `completed: Boolean` (true if the ride has ended).
>
> Source: Makura S.M (2025)

Help Jacob fulfil the following objectives:

(a) Write the `Ride` entity class. (5 Marks)

(b) Implement `addRide` method to add a new ride. (3 Marks)

(c) Provide a JPQL query to find all rides longer than 10km and completed, explain why JPQL helps here. (3 Marks)

(d) Write `listLongRides` method to list the long rides. (4 Marks)

Question 25 (15 Marks)

ORDBMS

Study the following scenario and answer the following questions:

> A retail company stores quarterly sales figures in an array column `sales_by_quarter` in the employees table. The database includes a predefined function `total_sales(sales INTEGER[])` that calculates the total annual sales from the array.
>
> You are asked to use this function and implement logic that ensures the `annual_sales` column is automatically updated whenever a new employee is inserted or their `sales_by_quarter` array is modified.
>
> Source: Makura S.M (2025)

Questions:

(a) Explain one advantage and one limitation of using arrays in object-relational databases. (4 Marks)

(b) Write a query that uses the `total_sales` function to display employee names and their total annual sales. (2 Marks)

(c) Complete the trigger function that automatically updates the `annual_sales` column whenever a new employee is inserted or their `sales_by_quarter` is updated. (4 Marks)

```
CREATE FUNCTION update_annual_sales() RETURNS _________ AS $$
BEGIN
    NEW.annual_sales := _____________(NEW.sales_by_quarter);
    RETURN ________;
END;
$$ _____________;
```

(d) Complete the trigger that calls the function before insert or update. (5 Marks)

```
CREATE _________ update_sales_trigger
___________ employees
FOR _________ OR _________
_________________ update_annual_sales();
```

Question 26 (10 Marks)

Semi-structured Data and XML

Study the following scenario and answer the following questions:

A company wants to validate their XML files containing employee records. Each record must include the following:

- `Name (string),`
- `Department (string),` and
- `Salary (decimal).`

They require that every employee must have a department and the salary must not be negative.

Source: Makura S.M (2025)

(a) Identify the XML schema type (simple, complex, or global) that is most appropriate for defining an employee element and justify your choice. (3 Marks)

(b) Write an XML Schema Definition (XSD) snippet for the employee element, enforcing the constraints mentioned above.(4 Marks)

Hint: You can use a structure similar to this:

```
<xs:element name="elementName">
   <xs:________>
      <xs:sequence>
         <!-- child elements go here -->
      </xs:sequence>
   </xs:__________>
</xs:element>
```

(c) Explain why XML Schema is preferred over DTD in this scenario. (3 Marks)

End of Semester Test Paper